SMART TRAFFIC MANAGEMENT USING VANET

Complete Python Simulation Project Report

Course: Ad Hoc Network (22ISE773)

Department of Information Science and Engineering

V2I communication, RSU-based traffic estimation and adaptive traffic signal control

Abstract

Traffic congestion at intersections increases travel time and waiting. This project models a four-way intersection in which vehicles periodically send V2I beacons to a Roadside Unit (RSU). The RSU estimates demand and an adaptive controller allocates green time between North-South and East-West approaches. A fixed-time baseline is also simulated. Metrics include average waiting time, queue length, throughput, packet delivery ratio and modelled V2I delay.

1. Introduction

A Vehicular Ad Hoc Network allows vehicles to communicate with other vehicles and roadside infrastructure. VANET applications include traffic monitoring, safety alerts and congestion management. This project uses connected-vehicle information to support adaptive signal timing.

2. Problem Statement

Fixed-time traffic signals do not adapt to changing traffic demand. Unequal traffic loads can cause avoidable queues. This project investigates whether vehicle-to-infrastructure traffic reports can help allocate green time according to current demand.

3. Objectives

Simulate vehicle arrivals and movement; model V2I beacon communication; estimate traffic demand at an RSU; allocate green time according to demand; and compare adaptive control with fixed-time control.

4. System Architecture

Vehicles generate traffic and periodically transmit beacons. The RSU receives beacons within its communication range, estimates demand by approach, and informs the adaptive signal controller. The controller adjusts the green-time split.

5. Simulation Assumptions

The model advances in one-second steps. Poisson arrival rates are North 0.14, South 0.10, East 0.40 and West 0.20 vehicles/second. Vehicle speed is modelled as 10 m/s, minimum spacing as 8 m, RSU range as 150 m, beacon interval as 5 seconds and packet-loss probability as 0.08. These are illustrative assumptions, not measured traffic counts.

6. V2I Communication

Vehicles within RSU range attempt periodic beacon transmission. Packet reception is probabilistic. Successfully received beacons are counted by approach and used as the RSU's latest traffic-demand estimate. A simplified 0.05-second delay is assigned to successfully received beacons. This is an abstract communication model, not a packet-level radio simulation.

7. Adaptive Signal Algorithm

A cycle uses a 54-second green budget and two 3-second amber periods. The controller allocates green time proportionally to the estimated North-South and East-West demand, subject to a minimum of 10 seconds and maximum of 44 seconds per phase. The fixed-time baseline uses 27 seconds per group.

8. Performance Metrics

Average waiting time is the mean elapsed time for vehicles that depart. Average queue length is the mean total queue per second. Throughput is vehicles served per minute. Packet Delivery Ratio is received beacons divided by attempted beacons. Average V2I delay is the modelled delay per received beacon.

9. Results and Discussion

Default run: 600 seconds, random seed 42. Adaptive VANET: 505 arrived, 432 served, average wait 49.73 s, average queue 44.48, maximum queue 74, PDR 92.43%, average V2I delay 0.050 s. Fixed-time: 505 arrived, 421 served, average wait 48.38 s, average queue 47.33, maximum queue 87, PDR 92.58%, average V2I delay 0.050 s. Adaptive control served more vehicles and had a lower average queue in this run, but average waiting time was slightly higher. This is a trade-off, not proof that adaptive control is always better.

10. Conclusion

The simulation demonstrates a VANET application in which V2I traffic reports support adaptive traffic signal timing. The fixed-time comparison provides a reproducible baseline under synthetic traffic conditions.

11. Limitations and Future Work

The project does not implement IEEE 802.11p/DSRC, a detailed MAC/PHY, calibrated radio propagation, GPS errors, lane changing or realistic car-following. Future work can integrate SUMO with Veins/OMNeT++ or NS-3, use measured traffic traces, and implement a detailed wireless MAC and routing model.

12. Syllabus Mapping

Unit 1: wireless communication concepts (abstracted). Unit 2: MAC protocol concepts motivate wireless access, but detailed MAC operation is not implemented. Unit 3: traffic monitoring, vehicle movement and V2I communication. Unit 4: dissemination of vehicle traffic information; routing protocols are not implemented. Unit 5: VANET applications, infrastructure use and vehicular network simulation.

Measured results – default 600-second run, seed 42

Metric	Adaptive VANET	Fixed-time
Vehicles arrived	505	505
Vehicles served	432	421
Average waiting time (s)	49.73	48.38
Average queue length	44.48	47.33
Maximum queue length	74	87
Packet delivery ratio	92.43%	92.58%
Average V2I delay (s)	0.050	0.050

All results are synthetic outputs of the stated model and are not field measurements.

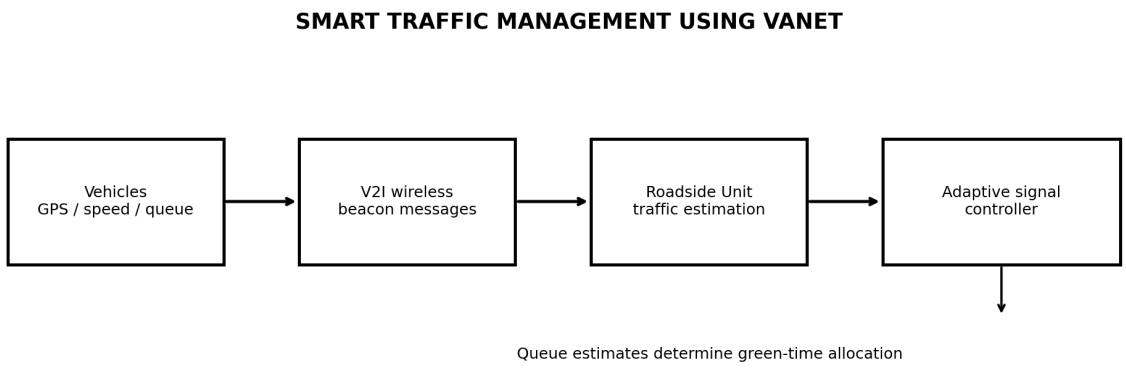


Figure 1. VANET system architecture

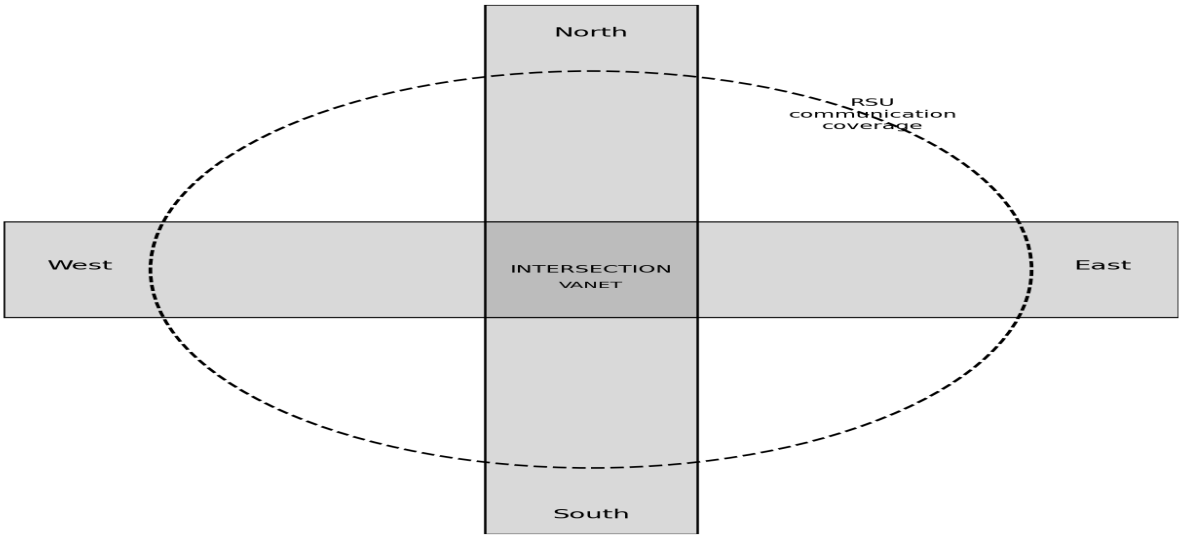


Figure 2. Four-way intersection and RSU communication coverage

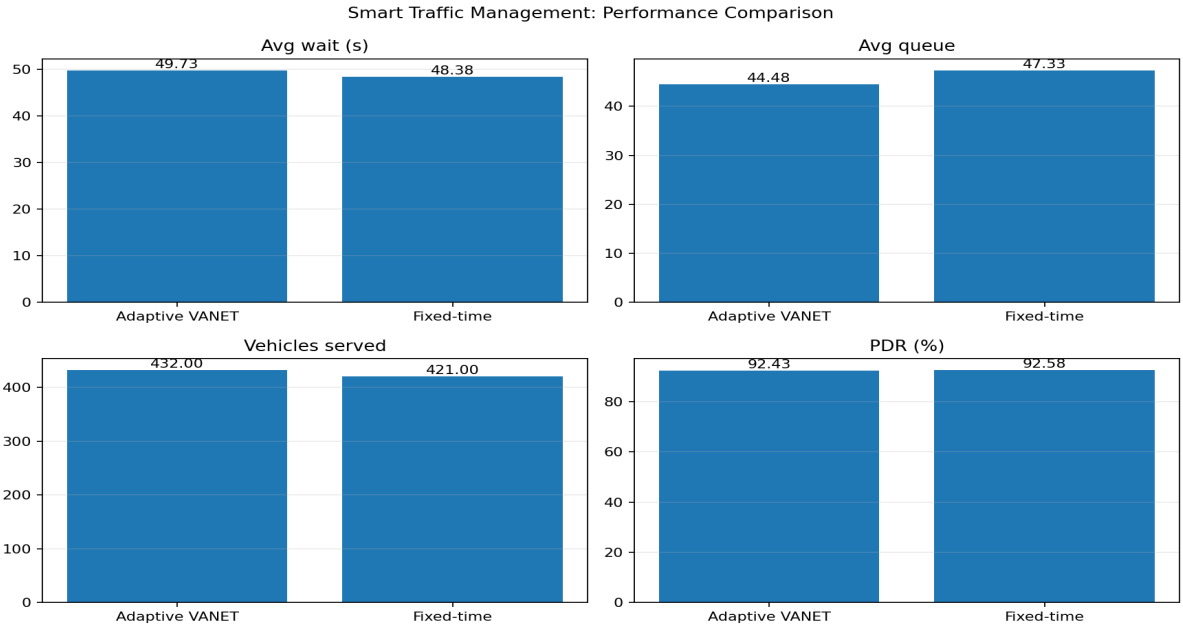


Figure 3. Adaptive versus fixed-time performance

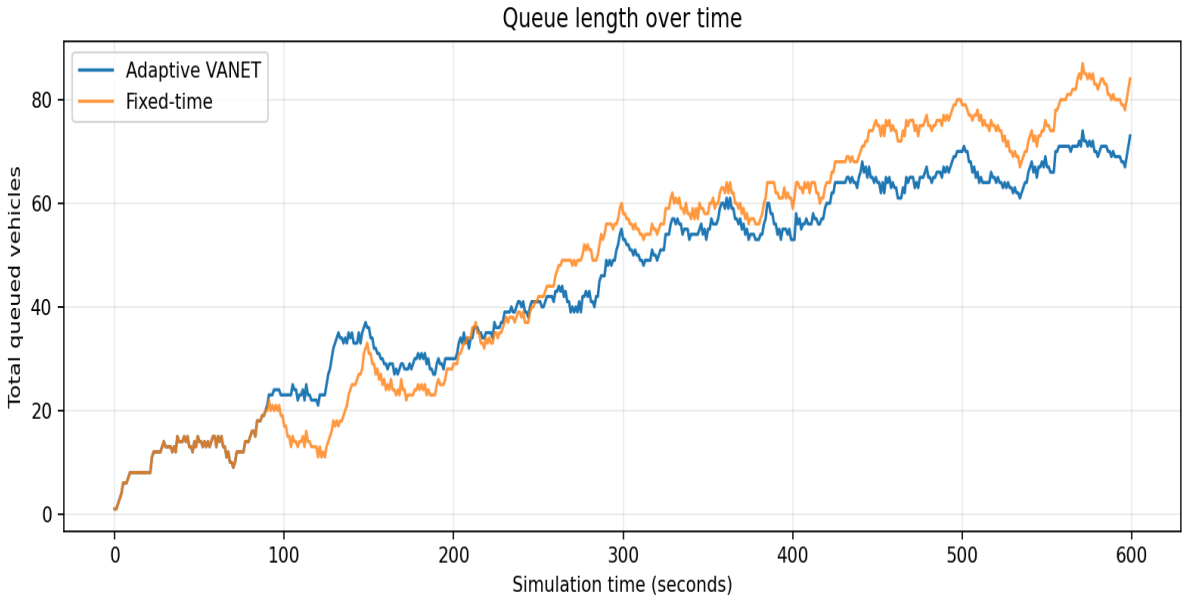


Figure 4. Queue length over simulation time